

CISC 327 - Assignment 1

Library Management – Role-Based Requirements

Librarian (Front-Desk Staff)

- The system shall allow front-desk staff to search for books by title or author.
- The system shall allow front-desk staff to search for library members by name.
- The system shall display whether a book is currently available or checked out.
- The system shall update a book's availability status when it is checked out or returned.
- The system shall allow front-desk staff to check books in and out.
- The system shall automatically inform front-desk staff when a member being served has an overdue book.
- The system shall support applying a small daily fine to overdue books.
- The system shall allow front-desk staff to complete common transactions quickly and using few steps.

Library Director

- The system shall provide reports showing how many books are currently checked out.
- The system shall provide reports identifying the most popular books.
- The system shall provide reports showing how many library members are active.
- The system shall provide measurable information that can be used to evaluate changes in late book returns.
- The system shall be affordable to operate and maintain within the library's limited budget.
- The system shall not require on-site IT staff for routine operation and maintenance.
- The system shall be easy to extend in the future to support features such as e-books or inter-library loans.

Library Member (Patron)

- The system shall allow library members to remotely check whether a book is currently available.

- The system shall allow library members to view their currently checked-out books and their due dates.
- The system shall allow library members to place a hold on a book that is currently checked out.
- The system shall send library members a reminder before a borrowed book reaches its due date.
- The system shall support a grace period for overdue books before an overdue consequence is applied.
- The system shall allow library members to access their library information using their library card number.
- The system shall allow a library member to renew a borrowed book when no other member is waiting for it.

Consolidated Quality-Check Worksheet

#	<u>Requirement</u>	<u>Source</u>	<u>Conflict</u>	<u>Ambiguity</u>	<u>Completeness</u>	<u>Consistency</u>	<u>Testability</u>	<u>Evolvability</u>	<u>Notes / Resolution</u>
1	The system shall allow front-desk staff to search for books by title or author.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	Clear search requirement. Can be tested by searching for known books using both a title and an author.
2	The system shall allow front-desk staff to search for library members by name.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	Clear and directly testable by searching for a known library member by name and verifying that the correct member is returned.
3	The system shall display whether a book is currently available or checked out.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	Clearly identifies the information the system must display. It can be tested by comparing the displayed status with a book's actual checkout status.
4	The system shall update a book's availability status when it is checked out or returned.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	Clearly identifies both events that trigger an availability update. It can be tested by checking a book out and returning it and verifying the status after each action.
5	The system shall allow front-desk staff to check books in and out.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	Clear front-desk functionality. Check-in and checkout can each be tested independently using a valid library member and book.

6	The system shall automatically inform front-desk staff when a member being served has an overdue book.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	The trigger and expected system behaviour are identified. It can be tested using members who have overdue books and members who do not.
7	The system shall support applying a small daily fine to overdue books.	Librarian	Fail	Fail	Fail	Pass	Fail	Pass	This conflicts with the Member's preference for a grace period rather than fines. "Small" does not define the amount of the fine, and it is unclear exactly when the daily fine begins. The conflict should remain open until the Director makes a final decision about the overdue policy.
8	The system shall allow front-desk staff to complete common transactions quickly and using few steps.	Librarian	Pass	Fail	Fail	Pass	Fail	Pass	"Quickly," "few steps," and "common transactions" are not precisely defined. The requirement should identify which transactions are included and establish measurable time or step limits before it can be objectively tested.
9	The system shall provide reports showing how many books are currently checked out.	Library Director	Pass	Pass	Pass	Pass	Pass	Pass	Clear reporting requirement. The report can be tested by comparing its number of checked-out books with the system's current checkout records.
10	The system shall provide reports identifying the most popular books.	Library Director	Pass	Fail	Fail	Pass	Fail	Pass	"Most popular" is not defined. Popularity could be based on checkouts, holds, or another measure. The measurement method and reporting period should be specified before this requirement is considered complete and testable.
11	The system shall provide reports showing how many library members are active.	Library Director	Pass	Fail	Fail	Pass	Fail	Pass	"Active" is not defined. The requirement needs a measurable definition of an active member and the period over which activity is measured before the reported number can be objectively tested.
12	The system shall provide measurable information that can be used to evaluate changes in late book	Library Director	Pass	Fail	Fail	Pass	Fail	Pass	The requirement does not specify what information will be measured or the time period used for comparison. The late-return metric and

	returns.								measurement period should be defined so the Director can determine whether late returns have increased or decreased.
13	The system shall be affordable to operate and maintain within the library's limited budget.	Library Director	Pass	Fail	Fail	Pass	Fail	Pass	"Affordable" and "limited budget" do not provide a measurable cost threshold. A maximum acceptable operating or maintenance cost would need to be established before this requirement can be objectively tested.
14	The system shall not require on-site IT staff for routine operation and maintenance.	Library Director	Pass	Fail	Fail	Pass	Fail	Pass	The overall intent is clear, but "routine operation and maintenance" is not defined. The requirement should specify which routine tasks library staff must be able to perform without assistance from on-site IT staff.
15	The system shall be easy to extend in the future to support features such as e-books or inter-library loans.	Library Director	Pass	Fail	Fail	Pass	Fail	Pass	"Easy to extend" is subjective and does not provide a measurable standard. The expected level of extensibility should be clarified while keeping e-books and inter-library loans outside the scope of the current system.
16	The system shall allow library members to remotely check whether a book is currently available.	Library Member	Pass	Pass	Pass	Pass	Pass	Pass	Clear member-facing requirement. It can be tested remotely using both an available book and a checked-out book and confirming that the correct status is displayed.
17	The system shall allow library members to view their currently checked-out books and their due dates.	Library Member	Pass	Pass	Pass	Pass	Pass	Pass	Clearly identifies the information members must be able to view. It can be tested by comparing the displayed books and due dates with the member's actual checkout records.
18	The system shall allow library members to place a hold on a book that is currently checked out.	Library Member	Pass	Pass	Pass	Pass	Pass	Pass	The basic hold functionality and condition are clearly stated. It can be tested by attempting to place a hold on a book that is currently checked out.
19	The system shall send library members a	Library Member	Pass	Fail	Fail	Pass	Fail	Pass	The requirement does not specify how long before the

	reminder before a borrowed book reaches its due date.								due date the reminder should be sent or how the reminder should be delivered. The timing and delivery method should be defined before it can be objectively tested.
20	The system shall support a grace period for overdue books before an overdue consequence is applied.	Library Member	Fail	Fail	Fail	Pass	Fail	Pass	This conflicts with the Librarian's preference for a daily fine on overdue books. The length of the grace period and the consequence after it expires are also unspecified. The policy should remain open until the Director makes a final decision.
21	The system shall allow library members to access their library information using their library card number.	Library Member	Pass	Fail	Fail	Pass	Fail	Pass	It is unclear whether the library card number alone authenticates the member or whether another credential is required. The authentication process should be clarified before this requirement can be completely implemented and tested.
22	The system shall allow a library member to renew a borrowed book when no other member is waiting for it.	Library Member	Pass	Pass	Pass	Pass	Pass	Pass	The renewal condition is clearly stated. It can be tested by attempting renewal when no other member is waiting and comparing that with a case where another member has placed a hold.

AI-Generated Requirement

#	Requirement	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes / Resolution
1	The system shall maintain a record of every book in the library's collection.	Pass	Pass	Pass	Pass	Pass	Pass	The requirement clearly identifies what information the system must maintain. It can be tested by adding books to the collection and verifying that each book has a corresponding system record.
2	The system shall maintain a record of which library member has borrowed each	Pass	Pass	Pass	Pass	Pass	Pass	The requirement clearly establishes the relationship between a checked-out book and the member who borrowed it. It can be tested

	checked-out book.							by checking out a book and verifying that the correct member is associated with it.
3	The system shall maintain lending records for a collection of approximately 3,000 books and 800 active members.	Pass	Fail	Fail	Pass	Fail	Pass	“Approximately” does not establish an exact capacity requirement, and “active member” is not defined in the scenario. A specific minimum capacity and definition of an active member would make the requirement measurable and testable.
4	The system shall provide a way for staff to identify books that have not been returned and may be missing.	Pass	Fail	Fail	Pass	Fail	Pass	“May be missing” is not clearly defined. The requirement does not specify when an unreturned book should be considered potentially missing. A specific rule or condition for identifying a missing book would be needed for objective testing.

Comparison of Group and AI-Generated Requirements

The AI-generated requirements captured some of the basic needs of the library, but the role-play produced a much more detailed understanding of what the stakeholders actually need. The biggest difference was that the AI did not identify the conflict surrounding overdue books and fines. Through the role-play, the Librarian supported a small daily fine because they believed it would encourage members to return books on time, while the Library Member preferred reminders and a grace period because they viewed fines as unfair. The Library Director had not made a final decision and was concerned about balancing timely returns with avoiding policies that could negatively affect lower-income patrons. This conflict could not have been discovered from the scenario alone because the scenario only states that the library wants to reduce late and missing books.

The role-play also surfaced many requirements that the AI missed. These included allowing members to place holds and renew books, providing due-date reminders, allowing members to view their borrowed books and due dates, using a library card number to access member information, automatically notifying staff about overdue books, and providing reports for the Library Director. The role-play also revealed concerns about ease of use, maintenance without on-site IT staff, and the ability to extend the system in the future. None of these details were available in the original scenario, which was intentionally incomplete.

The AI did make some assumptions when turning the scenario into requirements. For example, it stated that the system should maintain lending records for approximately 3,000 books and 800 active members. Although these numbers were provided in the scenario, the library never explicitly stated that they were system capacity requirements. The AI also proposed that staff should be able to identify books that “may be missing.” While reducing missing books was an explicit goal, the scenario never explained when a book should be classified as missing. This created ambiguity that would require additional stakeholder clarification.

There was still significant agreement between the two lists. Both recognized that the system needs to maintain information about books, track which members have borrowed them, and support staff in managing library information. These requirements align with the scenario's basic request for a system that keeps track of books and who has borrowed them. One useful requirement identified by the AI that our group had not explicitly included was maintaining a record for every book in the library's collection. Our role-play requirements focused heavily on searching, availability, borrowing, and returning books, but did not directly state that every library book must have a system record. This was a useful addition to consider for our final refined requirements.

Final Refined Requirements List

Based on the stakeholder role-play, quality-check process, and AI comparison, the following represents the group's final refined requirements. The AI's useful idea that every book should have a system record has been incorporated because it is directly grounded in the scenario's requirement to keep track of the library's books.

Book and Lending Management

- The system shall maintain a record of every book in the library's collection.
- The system shall maintain a record of the library member who has borrowed each checked-out book.
- The system shall allow front-desk staff to search for books by title or author.
- The system shall allow front-desk staff to search for library members by name.
- The system shall display whether a book is currently available or checked out.
- The system shall allow front-desk staff to check books in and out and shall update the book's availability status accordingly.

- The system shall automatically inform front-desk staff when a member being served has an overdue book.

Library Member Features

- The system shall allow library members to remotely check whether a book is currently available.
- The system shall allow library members to view their currently checked-out books and their due dates.
- The system shall allow library members to place a hold on a book that is currently checked out.
- The system shall allow a library member to renew a borrowed book when no other member is waiting for that book.
- The system shall send library members a reminder before a borrowed book reaches its due date.
- The system shall allow library members to access their library information using their library card number.

Reporting and Administration

- The system shall provide the Library Director with a report showing the number of books currently checked out.
- The system shall provide the Library Director with reports identifying popular books based on their checkout activity.
- The system shall provide the Library Director with reports showing library member activity.
- The system shall maintain information about late book returns that can be used to evaluate changes in the number of late returns over time.
- System Quality and Constraints
- The system shall support front-desk staff in completing common library transactions with minimal steps and delays.
- The system shall be operable and maintainable without requiring on-site IT staff for routine library operations.
- The system shall be designed to allow future capabilities, such as e-books and inter-library loans, to be added without requiring a complete redesign of the system.
- The system shall be designed with the library's limited operating and maintenance budget in mind.

Open Policy Requirement

- The system shall support a configurable overdue-book policy so that the library's final policy regarding grace periods and overdue consequences can be implemented once determined.

Open stakeholder conflict: The exact overdue-book policy remains unresolved. The Librarian supports a small daily fine, while the Library Member prefers reminders and a grace period rather than fines. The Library Director has not made a final decision and is leaning toward a small grace period followed by a modest fine for repeated lateness. Therefore, the final requirements do not impose a specific fine amount or grace-period length. This preserves the stakeholder conflict rather than making an unsupported policy decision.